AIMS
Airbus Material Specification

Aluminium alloy (7040)
Solution treated, controlled stretched and artificially aged (T7651)
High Toughness
Plate
120,0 mm £ a £ 145,0 mm
Close tolerance flatness

Material Specification

Published and distributed by :
AIRBUS S.A.S.
ENGINEERING DIRECTORATE
31707 BLAGNAC Cedex
FRANCE

Contents list

1 Scope

This Material Specification defines the requirements of aluminium alloy 7040 T7651 high toughness plate, for aerospace application.
Unless otherwise specified by the drawing, the order or the inspection schedule, the present specification shall be used in conjunction with the referenced documents.

2 Application

None

3 Normative references

This Airbus specification incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this Airbus specification only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred to shall be applied.

ABS 5324	Aerospace series, Close-tolerance flatness, Plates, Aluminium alloy 7040/7050, High Toughness, Close Tolerance Flatness, Thickness 75mm $\leq$ a $\leq$ 220mm Dimensions
AIMS 03-02-000	Technical specification - Aluminium and aluminium alloy wrought products - Plate
EN 6072	Aerospace series – Test methods - Constant amplitude fatigue testing
ABP 6-5230	Testing by Penetrant Flaw Detect.
ASTM E466	Standard Practice for Conducting Force Controlled Constant Amplitude Axial Fatigue Test of Metallic Material.
ASTM E561	Standard Practice for R-Curve Determination
ASTM E647	Standard Test Method for Measurement of Fatigue Crack Growth Rates
ASTM G34	Standard Test Method of exfoliation corrosion susceptibility in 2XXX and 7XXX series aluminium alloys (EXCO test)
EN 2032-2	Aerospace series - Metallic materials - Part 2 Coding of the metallurgical condition in delivery condition
EN 4500-2	Aerospace series - instructions for the drafting and use of metallic material standards-Part 2 - Specific requirements for aluminium, aluminium alloys and magnesium alloys.

4 Technical requirements

4.1 General

Specification values are individual minimums except where stated

4.2 Specific

The data requirements given in the tables are in accordance with the relevant material part of EN4500-2

5 Designation

Refer to ABS 5324B

6 Technical specification

The general technical requirements and responsibilities corresponding to this Material Specification are described in the Technical Specification AIMS 03-02-000.

			Si	Fe	Cu	Mn	Mg	Zn	Ti	Zr	Others Each	Others total	Al
1	Material designation		Aluminium alloy 7040										
2	Chemical Composition %	Element	Si	Fe	Cu	Mn	Mg	Zn	Ti	Zr	Each	total	Al
		min.	-	-	1,5	-	1,7	5,7		0,05	-	-	Rem
		max.	0,10	0,13	2,3	0,04	2,4	6,7	0,06	0,12	0,05	0,15	
3	Method of melting		-										
4	Form Method of production Limit dimension (mm)		Plate Rolled $120,0 \leq a \leq 145,0$										
5	Technical specification		AIMS 03-02-000										

6	6.1 Delivery condition and heat treatment	T7651 Solution treated + controlled stretched + artificially aged
	6.2 Delivery condition code	U [1]
7	Use condition and heat treatment	T7651 As delivered

8	Sample Test piece Heat treatment				T7651			
9	Dimension concerned		a	mm	$120,0 \leq a \leq 145,0$			
10	Thickness of cladding on each face			%				
11	Direction of test piece				L	LT	ST	ST45L [2]
12	TENSILE	Temperature	θ	°C				
13	TENSILE	Proof stress	$R_{p0,2}$	MPa	455	440	415	-
14	TENSILE	Strength	Rm	MPa	490	485	465	-
15	TENSILE	Elongation	A	%	6	4	3	1.6 [3]
16	TENSILE	Reduction of area	Z	%	-			
17	Hardness		HB	-	-			
18	Shear strength		Rc	MPa	-			
19	Bending		k	-	-			
20	Impact strength		k	-				
21	CREEP	Temperature	θ	°C	-			
22	CREEP	Time		h	-			
23	CREEP	Stress	σ_a	MPa	-			
24	CREEP	Elongation	a	%	-			
25	CREEP	Rupture stress	σ_R	MPa	-			
26	CREEP	Elongation at rupture	A	%	-			
27	Notes (see Line 98)				1), 2), 3), 4) and 5)			

<table>
<tr><td>32</td><td>Electrical conductivity</td><td>7</td><td colspan="5">γ ≥ 22,7 MS/m</td></tr>
<tr><td rowspan="2">39</td><td rowspan="2">Stress corrosion</td><td>6</td><td colspan="5">Test in ST direction, stress level 180MPa,
Constant load, length of exposure 30 days</td></tr>
<tr><td>7</td><td colspan="5">Criteria: no failure (30 days)</td></tr>
<tr><td rowspan="2">40</td><td rowspan="2">Fracture toughness (K_{1C})</td><td rowspan="2">7</td><td colspan="2">Dimension,
(mm)</td><td>L – T
(MPa√m)</td><td>T – L
(MPa√m)</td><td>S – L
(MPa√m)</td></tr>
<tr><td colspan="2">120,0 ≤ a ≤ 145,0</td><td>25</td><td>22</td><td>22</td></tr>
<tr><td rowspan="5">40</td><td rowspan="5">R-Curve (K_{co}) [2)]</td><td>1</td><td colspan="5">ASTM E561</td></tr>
<tr><td>2</td><td colspan="5">1 per lot</td></tr>
<tr><td>3</td><td colspan="5">CCT specimen, W = 160mm, 2ao/W = 0,25,
thickness = 6,3 mm, L-T direction, a/4</td></tr>
<tr><td rowspan="2">7</td><td colspan="5">The toughness shall not be lower than the following data</td></tr>
<tr><td colspan="2">120,0 ≤ a ≤ 145,0</td><td colspan="3">64</td></tr>
<tr><td>44</td><td>External defects</td><td>-</td><td colspan="5">See AIMS 03-02-000</td></tr>
<tr><td rowspan="8">46</td><td rowspan="8">Fatigue</td><td>3</td><td colspan="5">T type specimen in accordance with EN 6072</td></tr>
<tr><td rowspan="7">7</td><td colspan="5">The curve of the test results, calculated in accordance with EN 6072, shall not lie below the reference curve defined by the following data:</td></tr>
<tr><td>$\sigma_{max\ nett}$ (MPa)</td><td>No. of Cycles (N)</td><td>R</td><td>Kt</td><td>Direction</td></tr>
<tr><td>280</td><td>1,0E04</td><td rowspan="5">0,1</td><td rowspan="5">2,3</td><td rowspan="5">L-T</td></tr>
<tr><td>220</td><td>2,0E04</td></tr>
<tr><td>170</td><td>5,0E04</td></tr>
<tr><td>140</td><td>1,0E05</td></tr>
<tr><td>130</td><td>1,0E06</td></tr>
<tr><td rowspan="7">46</td><td rowspan="7">Fatigue (Smooth)</td><td>-</td><td colspan="5">See AIMS 03-02-000</td></tr>
<tr><td>2</td><td colspan="5">Two samples from each end of the parent plate</td></tr>
<tr><td>3</td><td colspan="5">Un-notched circular cross section specimen with tangentially blended fillets in accordance with ASTM E466 to be extracted from the LT direction.</td></tr>
<tr><td rowspan="4">7</td><td colspan="5">The specimen shall be tested at R=0.1, at a maximum stress of 240MPa and shall meet the following fatigue requirements</td></tr>
<tr><td colspan="5">Minimum individual life time: 120000 cycles</td></tr>
<tr><td colspan="5">Log average of 4 samples: 160000 cycles</td></tr>
<tr><td colspan="5">Runout: 300000 cycles</td></tr>
<tr><td rowspan="3">49</td><td rowspan="3">Exfoliation corrosion</td><td>1</td><td colspan="5">ASTM G34 [4)]</td></tr>
<tr><td>3</td><td colspan="5">a/2 position</td></tr>
<tr><td>7</td><td colspan="5">Criteria : Better or equal to grade EB of ASTM G34</td></tr>
</table>

61	Internal defects	-	See AIMS 03-02-000, plus tests indicated below.			
		1	Liquid Penetrant Testing in accordance with ABP 6-5230			
		2	Two samples from each end of the parent plate			
		3	LT-ST plate taken at b/2 at each end of plate			
		6	120,0mm ≤ a ≤ 145,0mm			
		7	The density of porosity shall be reported, including the distribution per size of voices and position through the thickness.			
69	Elastic modulus	3	L Direction			
		7	The value must be within the following range 68000 MPa ≤ E ≤ 72000 MPa 71000 MPa ≤ E_C ≤ 75000 MPa			
70	Compression Proof stress $R_{0,2C}$	7	Dimension (mm)	L direction (MPa)		
			120,0 ≤ a ≤ 145,0	445		
71	Crack propagation	1	ASTM E647			
		3	CCT Specimen, W=101mm, $2a_I$ =10mm, L=305mm, thickness=6.35mm. R=0,1. L-T, a/4 position, Test Conditions [5)]			
		7	The crack propagation rates shall not be greater than the following data:			
			ΔK MPa√m	da/dN (mm/cycle)	R	
			10	3.0E-04	0,1	
			15	1.0E-03		
			20	1.5E-03		
			25	3.0E-03		
			30	6.0E-03		
82	Batch uniformity	1	See AIMS 03-02-000			
		7	Electrical conductivity	T7651, γ = 23,7 MS/m (typical)		
				1,0 MS/m per product	1,5 MS/m per batch	
85	Bearing stress				120,0 ≤ a ≤ 145,0mm	
					L (MPa)	LT (MPa)
			e/D = 1,5	F_{bru}	720	750
				F_{bry}	630	630
			e/D = 2,0	F_{bru}	950	980
				F_{bry}	700	720

98	Notes		1) See EN 2032-2. 2) Extracted from t/4 location. Batch Release test. May be reduced to capability on agreement between manufacturer and responsible AIB partner. 3) Provisional values. These values will be reviewed by the manufacturer and responsible Airbus partner. The result of this review will lead to the confirmation or amendment of these values. 4) Test solution = 15 ± 1ml/cm^2 5) Test conditions: Room temperature (18-35°C), Relative humidity 35 - 50%, test frequency ≤ 30Hz.
99	Typical use	-	Spars
100	Qualification		Frequency of qualification testing is included in AIMS 03-02-000.

RECORD OF REVISIONS

Issue	Clause modified	Description of modification
1 01/04		New Standard